# Exercise 1: Worked solutions

Electrical Engineering I · Module 92315 · Exercise 1

From Fundamentals to AI Apocalypse · personal study guide

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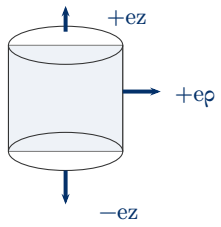

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Try every subpart before opening its answer. The steps show how the geometry determines the coordinates, how each integral is assembled, and how to check the result.

## Task 1: Cylinder in cylindrical coordinates

A right circular cylinder has radius $a$ and height $h$. Its axis is the $z$-axis, the origin is at the centre of the base, and positive $z$ points upward.

**Connected topic:** Flux and Gauss's law (§1.10) use closed surfaces and their outward normals to describe which way flux crosses a boundary.




**Three faces, three outward directions.** The disks have normals along the axis; the curved wall points radially outward. At the rims, two faces meet.

### 1a: Describe the three boundary surfaces

Try first: write the coordinate ranges for the base, curved side, and top.

*Answers: see "Task 1: Cylinder in cylindrical coordinates" under Answers at the end.*

### 1b: Give the outward unit normals

Try first: at each smooth face, which way points from the cylinder's interior into the surrounding space?

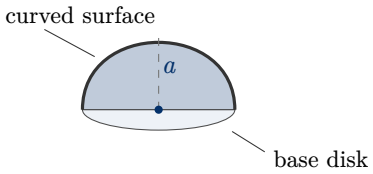
*Answers: see "Task 1: Cylinder in cylindrical coordinates" under Answers at the end.*

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## Task 2: Hemisphere of radius *a*

Take the upper half of a sphere and include its flat circular base in the total boundary area.

**Connected topic:** Spherical capacitors (§1.13) use spherical geometry. The guide's sphere illustrations help distinguish curved spherical surfaces from flat circular ones.




**Keep the boundary pieces separate.** The curved hemisphere contributes $2\pi a^2$; the flat base disk contributes $\pi a^2$.

### 2a: Total surface area

Try first: find the curved area and base area separately, then add the two pieces.

*Answers: see "Task 2: Hemisphere of radius a" under Answers at the end.*

### 2b: Volume

Try first: write the spherical volume element and integrate over the upper half-sphere.

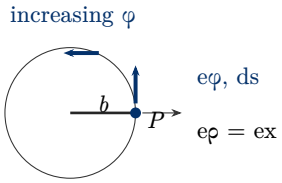
*Answers: see "Task 2: Hemisphere of radius a" under Answers at the end.*

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## Task 3: Field integral around a circle

For each supplied field, evaluate $\oint H \cdot \mathrm{d}s$ around a circle of radius $b$ centred at the origin. The exercise does not specify which way to traverse the loop, so assume increasing $\varphi$ from 0 to $2\pi$ (counterclockwise viewed from $+z$). Reversing the direction changes the sign of any nonzero answer.

**Connected topic:** Electrostatic line integrals (§1.8) use the same path-vector and dot-product steps. Here $H$ is a different supplied field; the connection is the integration method, not an identification of $H$ with $E$.




**At the right-hand point, the path tangent points up.** A radial or x-directed field points sideways there; an azimuthal field follows the tangent. Around the full loop, the x-directed contributions cancel.

### 3a: Uniform x-directed field

Given $H = H_0 e_x$. Try first: parameterize the loop and express its tangent in Cartesian components.

*Answers: see "Task 3: Field integral around a circle" under Answers at the end.*

### 3b: Radial field

Given $H = H_0 e_\rho$. Try first: compare the local field direction with the tangent at every point.

*Answers: see "Task 3: Field integral around a circle" under Answers at the end.*

### 3c: Azimuthal field growing with radius

Given $H = H_0(\rho/a) e_\varphi$. Try first: evaluate this field on the circle and compare it with $\mathrm{d}s$.

*Answers: see "Task 3: Field integral around a circle" under Answers at the end.*

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# Answers

## Task 1: Cylinder in cylindrical coordinates

### 1a · Surface coordinates

1. In cylindrical coordinates, $\rho$ measures distance from the axis, $\varphi$ gives the angle around it, and $z$ gives height. A point has position $r = \rho\, e_\rho + z\, e_z$.

   To describe a surface, hold one coordinate fixed and let the other two run over the part of the surface we want.

2. **Base:** every point is at height $z = 0$. The disk reaches from the axis to the wall, so $0 \le \rho \le a$, and it circles once with $0 \le \varphi < 2\pi$.

3. **Curved side:** every point is at radius $\rho = a$. Height runs from the base to the top, $0 \le z \le h$, while $0 \le \varphi < 2\pi$.

4. **Top:** every point is at height $z = h$. It is another full disk, so $0 \le \rho \le a$ and $0 \le \varphi < 2\pi$.

5. **Result:** base $(\rho, \varphi, 0)$; side $(a, \varphi, z)$; top $(\rho, \varphi, h)$, with the ranges above.

   **Check:** only the side fixes the radius. Each disk includes the centre and reaches all the way to $a$.

## Task 1: Cylinder in cylindrical coordinates

### 1b · Outward normals

1. A surface normal is perpendicular to the surface. "Outward" chooses between the two perpendicular directions by pointing away from the solid. Cylindrical coordinate unit vectors already have length 1.

2. At the base, the interior lies above the plane $z = 0$. The outward direction is therefore downward: $-\boldsymbol{e_z}$.

3. At the top, the interior lies below $z = h$. The outward direction is upward: $+\boldsymbol{e_z}$.

4. On the curved side, the solid occupies $\rho < a$. Increasing $\rho$ moves away from the axis and out of the cylinder, so the outward normal is $+\boldsymbol{e_\rho}$.

5. **Result:** base $-e_z$; side $+e_\rho$; top $+e_z$.

   **Check:** at either sharp rim, the base/top normal and side normal disagree. The rim has no single smooth-face normal; normals are defined face by face.

## Task 2: Hemisphere of radius a

### 2a · Curved surface plus base

1. Use spherical coordinates with polar angle $\theta$ measured down from $+z$. The curved hemisphere lies at $r = a$, with $0 \le \theta \le \pi/2$ and $0 \le \varphi < 2\pi$.
2. A small surface patch grows with both directions around the sphere. A change $\mathrm{d}\theta$ has length $a\,\mathrm{d}\theta$; a change $\mathrm{d}\varphi$ has length $a \sin\theta\,\mathrm{d}\varphi$. Thus $\mathrm{d}A = a^2 \sin\theta\,\mathrm{d}\theta\,\mathrm{d}\varphi$.
3. Integrate over the curved part: $A_{\text{curve}} = \int_0^{2\pi} \int_0^{\pi/2} a^2 \sin\theta\,\mathrm{d}\theta\,\mathrm{d}\varphi$.
4. The angular integral is $\int_0^{\pi/2} \sin\theta\,\mathrm{d}\theta = [-\cos\theta]_0^{\pi/2} = 1$. The azimuthal integral is $2\pi$, so $A_{\text{curve}} = 2\pi a^2$.
5. The flat base is a disk of radius $a$, with $A_{\text{base}} = \pi a^2$. Add the boundary pieces: $\boldsymbol{A_{\text{total}} = 2\pi a^2 + \pi a^2 = 3\pi a^2.}$

   **Check:** the curved half is half of $4\pi a^2$. The area has units of length squared. The shared circular edge has zero area, so it does not affect the sum.

## Task 2: Hemisphere of radius a

### 2b · Spherical volume integral

1. Inside the hemisphere, radius runs from the centre to the surface: $0 \le r \le a$. The angles cover $0 \le \theta \le \pi/2$ and $0 \le \varphi < 2\pi$.
2. The spherical-coordinate volume element is $\mathrm{d}V = r^2 \sin\theta\,\mathrm{d}r\,\mathrm{d}\theta\,\mathrm{d}\varphi$. The factor $r^2 \sin\theta$ is the coordinate Jacobian: small angular patches widen as we move away from the origin.
3. Set up the integral: $V = \int_0^{2\pi} \int_0^{\pi/2} \int_0^{a} r^2 \sin\theta\,\mathrm{d}r\,\mathrm{d}\theta\,\mathrm{d}\varphi$. Since the factors depend on separate variables, evaluate them separately.
4. $\int_0^{a} r^2\,\mathrm{d}r = a^3/3$; $\int_0^{\pi/2} \sin\theta\,\mathrm{d}\theta = 1$; $\int_0^{2\pi} \mathrm{d}\varphi = 2\pi$.
5. **Result:** $\boldsymbol{V = (a^3/3)(1)(2\pi) = 2\pi a^3/3.}$

   **Check:** it is half the full sphere's $4\pi a^3/3$ volume, and its units are length cubed.

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## Task 3: Field integral around a circle

### 3a · Uniform field

1. Parameterize the circle by its angle: $r(\varphi) = b \cos \varphi \; e_x + b \sin \varphi \; e_y$, with $0 \leq \varphi \leq 2\pi$.
2. Differentiate with respect to the path parameter. The line element is $\mathrm{d}s = (-b \sin \varphi \; e_x + b \cos \varphi \; e_y) \, \mathrm{d}\varphi = b \; e_\varphi \, \mathrm{d}\varphi$.
3. Take the dot product with the constant field: $H \cdot \mathrm{d}s = H_0 e_x \cdot b \; e_\varphi \, \mathrm{d}\varphi = -b \; H_0 \sin \varphi \, \mathrm{d}\varphi$.
4. Integrate over the whole circle: $\oint H \cdot \mathrm{d}s = -b \; H_0 \int_0^{2\pi} \sin \varphi \, \mathrm{d}\varphi = b \; H_0 [\cos \varphi]_0^{2\pi}$.
5. **Result:** $\oint \boldsymbol{H} \cdot \mathbf{d}\boldsymbol{s} = \mathbf{0}$. The field has a tangential component on parts of the loop, but the positive and negative contributions cancel over one full turn.

   **Check:** the zero is caused by cancellation around the loop, not by the field being perpendicular to the path at every point.

## Task 3: Field integral around a circle

### 3b · Radial field

1. Use the same circle parameterization. At radius $b$, the line element is $\mathrm{d}s = b \; e_\varphi \, \mathrm{d}\varphi$, tangent to the circle.
2. The radial unit vector $e_\rho$ points out from the origin, while $e_\varphi$ follows the circle. In cylindrical coordinates these perpendicular directions satisfy $e_\rho \cdot e_\varphi = 0$.
3. Therefore $H \cdot \mathrm{d}s = H_0 e_\rho \cdot b \; e_\varphi \, \mathrm{d}\varphi = 0$ at each point, so its integral is also zero.
4. **Result:** $\oint \boldsymbol{H} \cdot \mathbf{d}\boldsymbol{s} = \mathbf{0}$.

   **Check:** unlike 3a, the integrand is zero point by point. No cancellation between different parts of the circle is needed.

## Task 3: Field integral around a circle

### 3c · Tangential field

1. The path is still the circle $\rho = b$, so $\mathrm{d}s = b \; e_\varphi \, \mathrm{d}\varphi$, with $\varphi$ increasing around the loop.
2. Substitute $\rho = b$ into the field: $H = H_0(b/a)e_\varphi$ on this path. The field strength is constant along this particular circle.
3. Both vectors point along $e_\varphi$. Since this unit vector has dot product 1 with itself, $H \cdot \mathrm{d}s = H_0(b^2/a) \, \mathrm{d}\varphi$.
4. Integrate the constant around one revolution: $\oint H \cdot \mathrm{d}s = \int_0^{2\pi} H_0(b^2/a) \, \mathrm{d}\varphi = 2\pi H_0 b^2/a$.
5. **Result:** $\oint \boldsymbol{H} \cdot \mathbf{d}\boldsymbol{s} = \mathbf{2\pi H_0 b^2/a}$. It is positive for the assumed counterclockwise direction.

   **Check:** $H_0$ has units A/m, and $b^2/a$ has units m, so the integral has units A. Reversing the loop makes it negative.

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